

Neurophysiological encoding of aversive prediction errors

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Abstract

Aversive prediction error (PE) brain signals generated by unexpected pain or pain absence are crucial for learning to avoid future pain. Yet, the detailed neurophysiological origins of PE signaling remain unclear. In this study, we combined an instrumental pain avoidance task with computational modeling and magnetoencephalography to detect time-resolved activations underlying pain expectations and aversive PE signals in the human brain. The task entailed learning probabilistically changing cue-pain associations to avoid receiving a pain stimulus. We used an axiomatic approach to identify general aversive PE signals that encode the degree to which the outcome deviated from expectations. Our findings indicate that aversive PE signals are generated in the alpha band (8–12 Hz) by the midbrain/diencephalon, lateral orbitofrontal cortex, and ventrolateral prefrontal cortex approximately 150 milliseconds after outcome delivery. Moreover, alpha oscillations in these regions also encoded pain expectations before the outcome. We speculate that this may facilitate the rapid generation of PEs by allowing outcome-related nociceptive activity to be integrated with ongoing predictive signals. Finally, decisions to avoid pain recruited alpha oscillations in the anterior cingulate and dorsomedial prefrontal cortices, suggesting their active engagement in comparing predicted action values. Overall, our data reveal the rapid neurophysiological mechanisms underlying the generation of aversive PEs and subsequent decision-making.

Keywords: Pain-related learning, Aversive prediction error, Magnetoencephalography, Brain oscillations

The authors use an instrumental pain avoidance task and magnetoencephalography brain imaging to reveal evolution of pain-related learning with high temporal, spectral, and spatial resolution.

1. Introduction

Pain serves an important teaching function by alerting us to the potentially harmful consequences of our actions. In the reinforcement learning framework,^{42,43} this teaching function relies on the brain generating aversive prediction error (PE) signals that

mark the difference between predicted and experienced outcome, for instance the expected or unexpected presence or absence of pain. Neuroimaging studies^{10,26,29,39,43,44,55} have identified candidate regions of pain-related PEs, including the periaqueductal gray area (PAG), amygdala, orbitofrontal cortex, insula, and anterior cingulate cortex (ACC). Notably, hemodynamic activity in the PAG^{26,39} at outcome onset fulfills axiomatic requirements^{39,40} for an aversive PE, including (1) enhanced activity for the presence vs absence of pain, regardless of expectations; (2) enhanced activity for low vs high pain expectations, regardless of outcome. Signals fulfilling these 2 axioms reflect an aversively signed PE, ranging from surprising pain (greatest aversion) to surprising pain avoidance (least aversion).

Because aversive PE signals are likely to be generated immediately after experiencing or avoiding a painful outcome, it is crucial to study them with high temporal precision. However, previous fMRI^{26,39} and EEG^{34,46} studies, respectively, lacked the temporal and spatial sensitivity to identify the rapid dynamics and anatomical origins of aversive PEs. Here, we leveraged magnetoencephalography (MEG)⁸ combined with an instrumental pain avoidance task^{26,39} to study aversive PEs and subsequent decision-making. To identify PE signaling, we examined the brain responses time-locked to the onset of outcome that are widely regarded to carry pain-related sensory information.³⁴ However, because aversive PEs result from the integration of sensory information with prior predictions, we further examined oscillatory activity in the alpha-beta (8–30 Hz) and gamma frequency (>30 Hz) that are not time-locked to the outcome. Indeed, the alpha-beta range of neurophysiological activity contributes to encoding expectancy effects mediated by feedback circuitry, whereas the

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gamma band encodes PEs through a feedforward mechanism.^{3,37}

However, in the case of aversive PEs, previous EEG data^{34,46,51,53} present evidence both in favor^{14,46} and against^{11,34,51} a role for oscillatory brain activity in PE encoding. This apparent discrepancy might stem from differences between studies in terms of learning paradigms (eg, classical vs instrumental), the types of stimuli used (eg, laser vs thermal), and whether they investigated signed (indicating worse-than or better-than expected outcomes) or absolute PE signals (indicating surprise irrespective of outcome valence). In the present study, we were particularly interested in aversively signed PEs that indicate worse-than-expected outcome because they are critically important for guiding behavior. Aversive PEs should peak in response to unexpected pain—signaling the need to revise action-outcome associations and avoid repeating the same action—and be minimal when pain is unexpectedly absent, reinforcing current behavior. Finally, we also explored how PEs influence future decisions via their effects on pain expectations—focusing specifically on brain activity during high-choice-difficulty trials, when available options have similar pain probabilities. Overall, this study offers a comprehensive examination of the spatial and temporal neurophysiological processes involved in pain avoidance learning, encompassing the generation of PEs to decision-making.

2. Methods

2.1. Participants

Thirty-three healthy controls (19–61 years old; 17 females and 16 males with a mean age of 33 ± 11 years and 30 ± 5 years, respectively) took part in the study. All participants met the inclusion criteria with no history of psychiatric, neurological, or pain disorders. The study was approved by the Cleveland Clinic Institutional Review Board, and all participants provided written informed consent. The study was registered in Clinicaltrials.gov (NCT04603417). Participants received compensation at a fixed amount of \$200 for completing magnetoencephalography (MEG) data collection and various questionnaires. In addition, participants who completed a T1w MRI received a \$50 compensation. Two participants were excluded from the study because they terminated participation before completing data collection. Two participants were excluded because of incomplete datasets due to technical issues (one involving the pressure pain device malfunction and another involving missing triggers needed for data analysis). Two more participants who completed the task fully were excluded because of poor task performance (see “Instrumental pain avoidance task”). In total, 6 participants were excluded. Thus, the final analyses included 27 participants as described in the Results section.

2.2. Instrumental pain avoidance task

The instrumental pain avoidance task (**Fig. 1A**) contained 150 trials. On each trial, participants made a choice between 2 visual objects (a diamond and a circle, or a star and a pentagon) during which they learned to select the option with the lowest probability of being followed by a painful stimulus. The probabilities associated with each option were independent of one another and drifted from trial to trial according to a pair of random walks (**Fig. 1B**). We used 2 different pairs of random walks with the criterion that they must cross (probabilistically reverse) at least twice. Each participant was randomly administered one of the 2 random walks.

The participants started out by trial and error, but soon learned cue-pain contingencies to avoid the option associated with high

probability of being followed by pain stimuli. Each trial started with the presentation of the 2 options, randomly displayed to the left or right of the screen for 1800 milliseconds. The participants entered their decision by clicking the left or right button of a mouse to select the option on the left or right side of the screen, respectively. Inactivity exceeding 2 seconds resulted in the computer randomly selecting a response. The chosen option was displayed for 200 milliseconds followed by an anticipatory period of 2 seconds during which a white crosshair was presented on the screen. Then the outcome—a brief painful pressure stimuli applied to participant’s upper extremity for 1300 milliseconds (see “Pressure pain”) or no-stimulus—was presented. Outcome onset was accompanied by a change of the crosshair to a central asterisk displayed for a period of 3 seconds. After the outcome, a visual analog scale (VAS) appeared on the screen for 5 seconds, and participants were instructed to rate their perceived pain by moving a slider on the computer screen using the mouse, only when their choice resulted in a pain outcome. The next trial started after a jittered intertrial interval of 1–3 seconds.

The full set of 150 trials (lasting ~33 minutes) was administered twice in each participant. The 2 trial sets differed by the site of pain delivery, which was either to the left hand (LH) or right hand (RH) thumb nail bed. The trial sets were separated by a short break lasting 5 to 10 minutes. The order of the pain delivery site between the first and second trial set was counterbalanced across participants, with 15 participants receiving pain delivered to the RH during the first trial set and LH during the second trial set. We used this task design to identify brain correlates of aversive PEs that are independent of the location and laterality of pain. When pain was delivered to RH, participants performed choice selections using a mouse on the LH and vice versa. Although the same random walk process was used for both trial sets, the random draws and choice objects were different.

Before data collection, on the same day, participants performed a practice session with limited number of trials, different pair of random walks, and different visual objects (rectangle and triangle) from the ones they received during the test session. During the practice session, instead of actual outcomes, the word “Pain” and “No Pain” appeared on the screen. During practice, participants were carefully instructed about all aspects of the experiment, except the actual probabilities of pain they would receive during the test session.

2.3. Pressure pain

Pressure pain stimuli was delivered to the participants thumb nailbed using a computer controlled pneumatic device (Integrated Instrument Development Facility, University of Colorado, Boulder) capable of transmitting controlled pressure to a piston surface with 0.397 cm^2 . Each stimulus lasted for 1300 milliseconds. Participants held on to a handle and inserted their thumb into a groove with a snugly fit piston on top. Pressure levels were individually titrated to be at a level 80 on a continuous pain rating scale ranging from 0 to 100 (0, no sensation; 10, nonpainful; 20, low pain; 50, moderate pain; 80, maximum tolerable pain; 100—excruciating pain) before the MEG session. Pressure levels were always set between 1 and 9 kg/cm^2 , and never can exceed 10 kg/cm^2 , which is the upper limit of the device output. Because of transmission delay associated with the pneumatic system, the piston movement in the pressure pain device occurred 35 milliseconds after outcome onset (change of the crosshair to a central asterisk in the task). Hence, all analyses were performed, and times noted are in relation to piston movement, that is, outcome delivery.

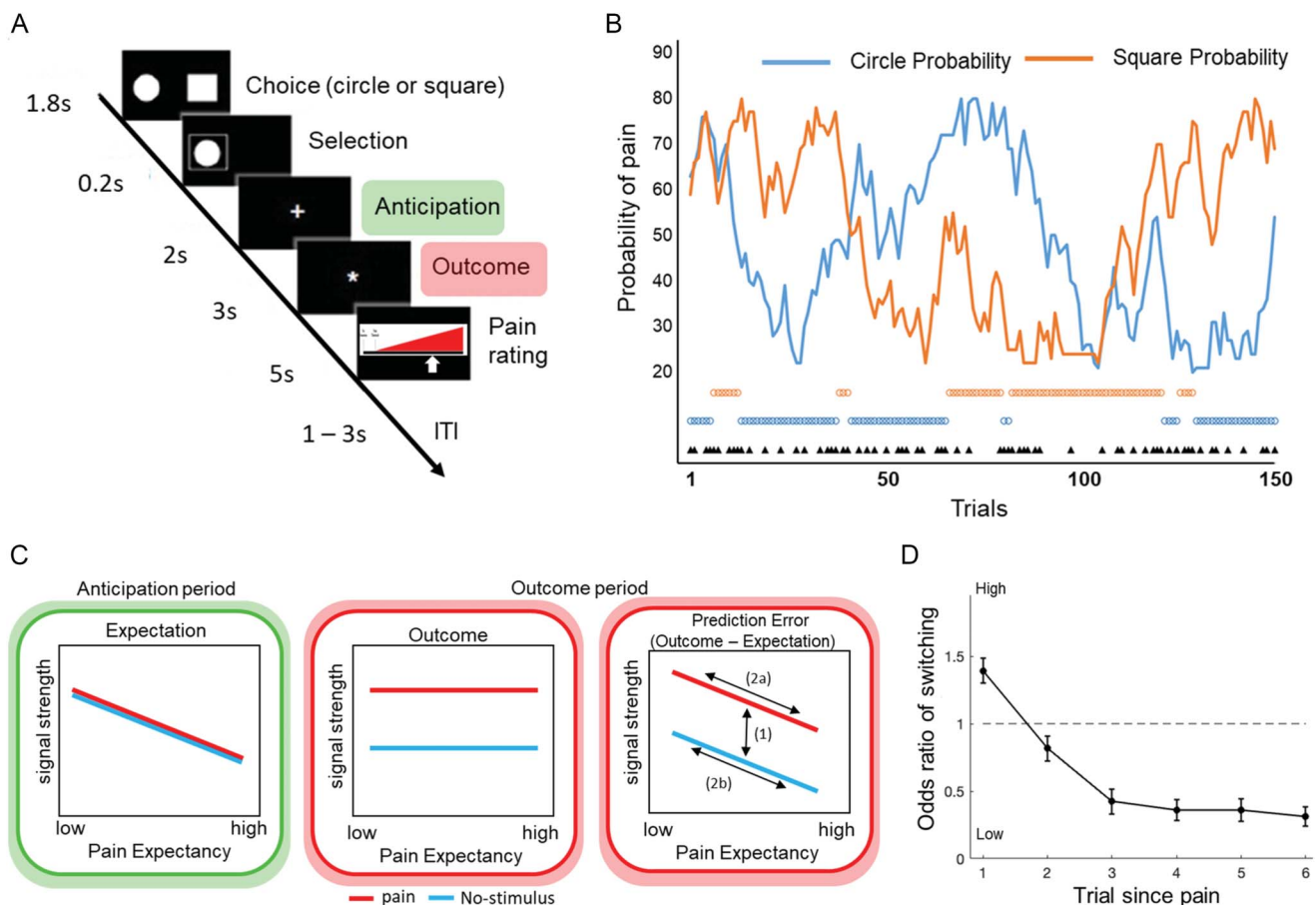


Figure 1. Instrumental pain avoidance task and behavioral results. (A) One experimental trial. Participants had 1.8 seconds to choose between 2 visual objects (using mouse clicks) associated with different probabilities of receiving a pain or no-stimulus outcome. Their choice was confirmed with a display of the selected object that lasted another 0.2 seconds. After an expectation (anticipatory) period of 2 seconds marked by a "+" symbol displayed on the screen, participants either received a painful stimulus or no-stimulus. The outcome (pain/no-stimuli) period was marked by a "*" asterisk fixation-point on the screen. If pain outcome was received, participants rated their pain using a visual analog scale. Trials were separated by a 1- to 3-second intertrial interval (ITI). (B) Random walk and choice selection from one participant. The blue and orange lines depict the probability of pain associated with each visual item over the 150 trials (one of 2 possible pairs of random walks used in the study). At the bottom of the curves, blue and orange dots represent the choices made by the participant, and black triangles represent pain outcomes. (C) Hypothesized aversive PE signal. PEs represent the difference between outcome and expectation. Left panel: Expectations preceding outcome delivery. Middle panel: Pain outcome is more aversive than no-stimulus outcome. Right panel: PEs linearly encode the relative aversiveness of the outcome, from the least aversive (no-stimulus when pain was expected) to the most aversive (pain when no-stimulus was expected). Therefore, axioms for an aversive prediction include (1) higher activity for pain vs no-stimulus outcome, regardless of expectations (axiom#1), and (2) higher activity for low vs high pain expectancy, regardless of outcome (axioms#2a-expectancy effects within pain trials and 2b-expectancy effects within no-stimulus trials). (D) Logistic regression model results. Probability of switching to the other possible choice as a function of pain received 1 to 6 trials back shown as mean odds ratio of both trial sets. Error bars represent standard error of the mean. Odds ratio > 1 indicates high likelihood of switching.

2.4. Magnetoencephalography data acquisition

A 306-channel MEG system (TRIUX neo, MEGIN Oy, Finland) that comprised 102 identical triple-sensor elements (2 orthogonal planar gradiometers and 1 magnetometer) was used to record neural activity while the participants performed the pain avoidance task inside a magnetically shielded room (Imedco, Switzerland). The height of the chair of the MEG device was adjusted so that the vertex of the scalp was in contact with the internal surface of the MEG apparatus, while the subjects were upright comfortably seated. A screen placed in the center of the field of view and approximately 1.5 m distance displayed the visual cues relayed by the E-Prime presentation software (Psychology Software Tools, Pittsburg, PA) and associated hardware.

The head position was continuously monitored by 5 head position indicator (HPI) coils affixed to the subject's scalp. Fiducial points (nasion, right and left auricular), along with head surface points and HPI locations collected using a Fastrack digitizer (Polhemus, Colchester, VT), allowed for

further registration of the MEG and MRI coordinate systems (see Magnetoencephalography data analyses). In addition to HPI coils, a standard 10-20 EEG channel montage was affixed to the participant's scalp. A pair of Ag-AgCl electrodes was taped across the participant's chest to capture the electrocardiography (EKG) synchronized with the MEG recording (data not reported). On the same hand where pain was delivered, skin conductance responses were measured using a pair of 2 gelled electrodes positioned on the palm (data not reported). Subjects were monitored continuously with a video camera to ensure alertness and continued attention to the task. Magnetoencephalography recordings were acquired continuously during each trial set of the task.

2.5. Reinforcement learning model-based analysis

Participants' decisions in the task were modeled as a function of previous choices and outcomes using a temporal difference model. The pain expectancy or expected probability of pain Q for the selected choice (eg, circle or square in **Fig. 1A**) in response

to the experienced outcome O (pain or no-stimulus) was updated on each trial t according to the following equation.

$$Q_{s,t+1} = Q_{s,t} + \alpha(O_t - Q_{s,t})$$

where $(O_t - Q_{s,t})$ reflect the prediction error (PE) on each trial. Outcome O was coded as -1 for pain and 0 for no-stimulus outcomes. Initial Q value of both choices was set to -0.5 . The value of the unchosen choice was not updated. Learning-rate parameter α controlled the extent to which Q values were updated in response to new outcome. Higher values of α resulted in faster updating. We used separate learning rates for pain and no-stimulus outcomes, α_{pain} and α_{nopain} , respectively. α_{pain} reflects how past pain outcomes influence future expectations, whereas α_{nopain} accounts for how past no-stimulus outcomes influence future expectations.

The probability P of choosing one choice s_1 over another s_2 at trial t was determined by a SoftMax distribution, where the inverse temperature parameter β controls the strength of the competition between the 2 options.

$$P_{s_1,t+1} = \frac{e^{Q_{s_1,t} \times \beta}}{e^{Q_{s_1,t} \times \beta} + e^{Q_{s_2,t} \times \beta}}$$

Inverse-temperature parameter β controlled the sensitivity of choice probability P to differences in Q values. If $\beta = 0$, both stimuli are equally likely to be chosen, irrespective of their expected pain probabilities. As β increases, the probability that the model chooses the choice with lower expected pain probability increases.

Model parameters α and β were estimated using a hierarchical Bayesian inference through Markov chain Monte Carlo (MCMC) sampling.¹ The methodology is explained elsewhere²⁶ in greater detail. In short, models were fitted hierarchically, such that the model parameters of each participant were sampled from a group-level distribution (hyperparameters), which were also assigned prior distributions (hyperpriors). For example, separate group-level parameters were estimated for first and second trial sets. To test for differences in the first vs second trial sets, we compared the posterior distributions of the group-level means.

We used weakly informative priors. The group-level distributions for all parameters were assumed to be normal distributions. We used a normal distribution with mean 0 and SD 1 and a positive half-Cauchy distribution with location 0 and scale 5 as hyperpriors for mean and SD hyperparameters, respectively, of these group-level distributions. We transformed the α parameters' unconstrained values to a $[0, 1]$ range using the inverse probit transformation² and β to a $[0, 20]$ range by multiplying its inverse-probit transformed values by 20.¹ We ran 4 independent MCMC chains with different starting values to generate 5000 posterior samples. After discarding first 1000 iterations of each chain as burn-in, we used every fifth iteration (to remove autocorrelation) to obtain 3200 representative samples (800 per chain) per parameter per model fit. All chains showed proper convergence with $R_{\text{hat}} < 1.1$.

In line with previous studies,^{26,39} we applied the mean model parameters (the individual-level posterior medians, averaged across all participants) to each participant's task data, that is, sequence of choices and outcomes to derive trial-wise pain expectancy estimates and PEs. The model provided signed PE estimates, that is, negative PEs for pain outcomes and positive PEs for no-stimulus outcomes. These estimates were used as parametric regressors on neurophysiological time series obtained from MEG source mapping, as detailed below.

2.6. Axiomatic analysis

Central to our approach are the axiomatic criteria used to identify the aversively signed PE signals. Contrary to the question of how expectations influence pain processing, which has been addressed in previous studies,^{11,14,46} we set out to answer the question of how brain learns to avoid pain. Addressing this question requires incorporating both pain and no-stimulus outcomes into the definition of PE, as both are necessary for understanding how actions are reinforced or adjusted based on their consequences. The axiomatic approach provides a valuable complement to correlational analyses between brain activity and computational estimates of PEs, to identify neural signals that genuinely reflect PEs. A key limitation of relying solely on correlations with computational PE estimates is that such estimates may, in practice, reflect only outcome (eg, pain vs no stimulus), or only expectations—but not their interaction, which is central to the definition of a PE in reinforcement learning theory.^{42,43} This arises because the standard computational formula for PEs (PE = outcome – expectation), by definition, positively correlates with outcome and negatively correlates with expectation. Consequently, neural signals responding to the presence of pain, regardless of expectations, or responding to expectations, regardless of the presence of pain, may significantly correlate with computational estimates of PEs without necessarily reflecting a PE signal.

The axiomatic framework^{12,40} provides a principled way to verify that a PE signal is not driven solely by outcomes or expectations, but by their interaction (**Fig. 1C**). In this study, we aimed to identify neural signals that track *aversively signed* PEs—that is, signals encoding how much worse an outcome is compared to expectations, along a continuum from the best case (unexpected pain avoidance) to the worst case (unexpected pain). Such a signal is central to adaptive behavior because it reinforces actions that avoided pain and penalizes those that did not. We define an aversively signed PE to simultaneously reflect 2 properties: it should be higher for pain outcomes than for no-stimulus outcomes (*axiom#1*, **Fig. 1C**, right panel), and higher when pain is less expected (*axioms#2a and 2b*, **Fig. 1C**, right panel). This is because unexpected pain is worse than expected pain, and unexpected relief is better than expected relief. For example, avoiding pain expected to be unavoidable is more rewarding than avoiding pain expected to be avoidable. Reversing the sign of both outcome and expectation would yield an *appetively signed* PE (eg, for rewards).

The rationale for using the axiomatic test for PEs is further detailed in Rutledge et al.⁴⁰ and has been applied to pain research by Roy et al.³⁹ and Jepma et al.²⁶ The current MEG study allowed us to examine the temporal dynamics of PE signals—specifically, how the brain compares outcomes to expectations over time. Given that expectations naturally precede outcomes, we also examined brain activity during the anticipation phase. In line with *axioms#2a and 2b* (**Fig. 1C**), we expected anticipatory signals to be stronger when pain was less expected (ie, low > high pain expectation). When such anticipatory signals also differentiate outcome in a manner consistent with *axiom#1* (pain > no-stimulus), this provides a plausible neural mechanism capable of rapidly generating aversively signed PEs—by integrating incoming pain-related information with preestablished expectations.

2.7. Magnetoencephalography data analyses

All MEG data were collected at 5000 (0.1–1660 Hz) samples/sec. Raw MEG data were preprocessed using tSSS algorithm⁵⁰ as

implemented in MaxFilter to remove internal and external artifacts with head movement estimation option (see Head movement compensation). All initial data preprocessing and analysis was performed using the Brainstorm toolbox for EEG/MEG analysis.⁴⁷ Physiological artifacts including eyeblinks, eye movements, and heartbeats were removed using a signal-space projection method. Data trials time-locked to the event of interest were resampled to 200 samples/sec and epoched making sure the trial duration was long enough to capture event-related brain responses and there was enough temporal padding to accommodate edge artifacts because of subsequent filtering or time-frequency decomposition. For PE analysis, MEG data were time-locked to outcome onset and epoched with a time window [−1500, 1500] milliseconds around each event. Time period preceding the onset of each trial, that is, display of 2 visual objects [−1000, 0] milliseconds was used as a baseline period for all analysis to avoid task related effects. Data from 204 planar gradiometers were chosen for all analysis because of their sensitivity to activity from local sources and improved signal to noise ratio. Single trials were inspected visually to make sure they were not contaminated with SQUID jumps or uncontrolled environmental noise. For evoked analysis, data trials were band-pass filtered between 1 and 30 Hz (linear phase FIR filter based on Kaiser window with 40 dB stop band attenuation). For oscillatory time-frequency (TF) analysis, data were only filtered using a notch filter (zero phase lag second order IIR filter with 3 dB notch bandwidth of 1 Hz) to remove 60-Hz line noise. All further analyses were performed on single trial data, except for visualizing MEG event-related fields and oscillatory TF responses (Figs. 2A and 3A), which were obtained by averaging trials.

The source or anatomical model was generated from T1-weighted magnetic resonance imaging (MRI) images of individual subject's head. The T1 scans were segmented into white and gray matter surfaces using Freesurfer,^{15,19} and cortical surfaces were tessellated into 5000 sources. The anatomical MRI and the reconstructed cortical surface were coregistered to MEG data by identifying the same fiducials and matching them with those collected during MEG session. In addition, hundreds of surface points were used to fine-tune this registration. In our cohort, we could not collect MRI data in one participant. Hence, for this participant data, we created a surrogate source model by warping the standardized Montreal Neurological Institute (MNI) MRI brain atlas to fit the subject's digitized scalp surface points. Then, a forward model (leadfield matrix) for the 5000 cortical sources was computed for each subject using the initial head position separately for each trial set, using the overlapping spheres head model.²³ Sensor level data trials were transformed to source estimates using the weighted minimum norm estimate (wMNE) approach with default parameter settings suggested for regularization and depth weighting in Brainstorm. wMNE current density estimates were further standardized with respect to noise covariance (computed using data from the baseline period) using dynamical Statistical Parametric Mapping approach dSPM.¹⁶ At each source, 3 time-series pertaining to x, y, and z directions were computed separately. The unconstrained source time-series were then flattened to compute the norm of the 3 orientations to yield one time-series per source. Further, source time-series from individual participants were projected on to the default MNI brain template for anatomical standardization.

For evoked analysis, spatial smoothing using a Gaussian kernel (FWHM = 3 mm) was performed on the resulting source time-series to further reduce interindividual variability. The absolute value of the estimated source evoked time-series (source space × time) for each trial from [0, 1000] milliseconds w.r.t outcome delivery was

time-averaged in steps of 35 milliseconds and exported to surface source GIFTI (.gii) files for further processing in SPM12³¹ (see Statistical Analyses). For oscillatory (TF) analysis, source time-series were subjected to a complex Morlet wavelet decomposition (central frequency = 1 Hz and FWHM = 3 seconds) with continuous time bins and frequency grouped into canonical bands (alpha [8, 12] Hz, beta [13, 30] Hz, low gamma [30, 60] Hz, and high gamma [61, 90 Hz]). The source TF maps (source space × time × frequency) were further normalized by performing a decibel (dB) conversion at each frequency band that is, taking the log of the ratio of total power value with respect to the mean total power over the baseline period (defined as the 1-second period before the appearance of 2 visual objects—refer to Fig. 1A). The absolute value of the estimated source TF maps for each trial and frequency band from [0, 1000] milliseconds w.r.t outcome delivery was time-averaged in steps of 35 milliseconds and exported to surface source GIFTI (.gii) files for further processing in SPM12.³¹

2.8. Head movement compensation

The estimated head movements from 6 quaternion data channels define 3 head rotations and 3 translations with respect to initial head position from the data set. Using this information, the position and orientation of the circumcenter with respect to the x-, y-, and z-axis were computed.⁴⁵ Further, these estimates were demeaned to obtain the normalized deviants, that is, translations and rotations from the average head position/orientation during the entire data set. Trial-by-trial estimates of head position were computed in the time window [−1500, 1500] milliseconds around outcome onset. To account for the nonlinear effects of head motion on the signal, we also computed the squares, cubes, and all their derivatives of the head movement parameters (resulting in a total of 36 estimates plus one constant), which were used as additional nuisance regressors in the Generalized Linear Model (GLM) analysis described below.

2.9. Statistical analyses

For all the analyses, we implemented a multi-level GLM approach where data from individual subjects were modeled at the first level, and the contrast image from each subject was entered into a group analysis at the second level. At the first level, the GLM design matrix columns had an intercept (mean), model-derived regressors of interest (described below for each analysis), and 37 head movement-related regressors of noninterest, corresponding to 150 trials in the task. At the second level, design matrix had a column of all 1's to evaluate group effect. In addition, sex (male = 1, female = −1) was entered as a second-level covariate to assess sex differences in our findings. The source t-maps resulting from the GLM analysis were subjected to a cluster-extent based thresholding methodology to identify significantly correlated sources. Cluster-extent based thresholding, a commonly used statistical correction procedure in neuroimaging consists of 2 steps.⁵⁴ (1) An initial cluster-defining threshold defines clusters by retaining groups of suprathreshold source signals. This initial threshold was set at a *P*-value of 0.01 for all analyses (except evoked analysis discussed below). (2) A cluster extent threshold that controls false discovery rate (FDR) determined (using random field theory²⁰) based on the estimated distribution of cluster sizes under the null hypothesis of no signal in any source in that cluster. The cluster-extent threshold was set at a *P*-value of 0.05 for all analyses. All analyses were performed separately for LH and RH trial sets (see Supplementary Material). We then performed a second-level conjunction between LH and

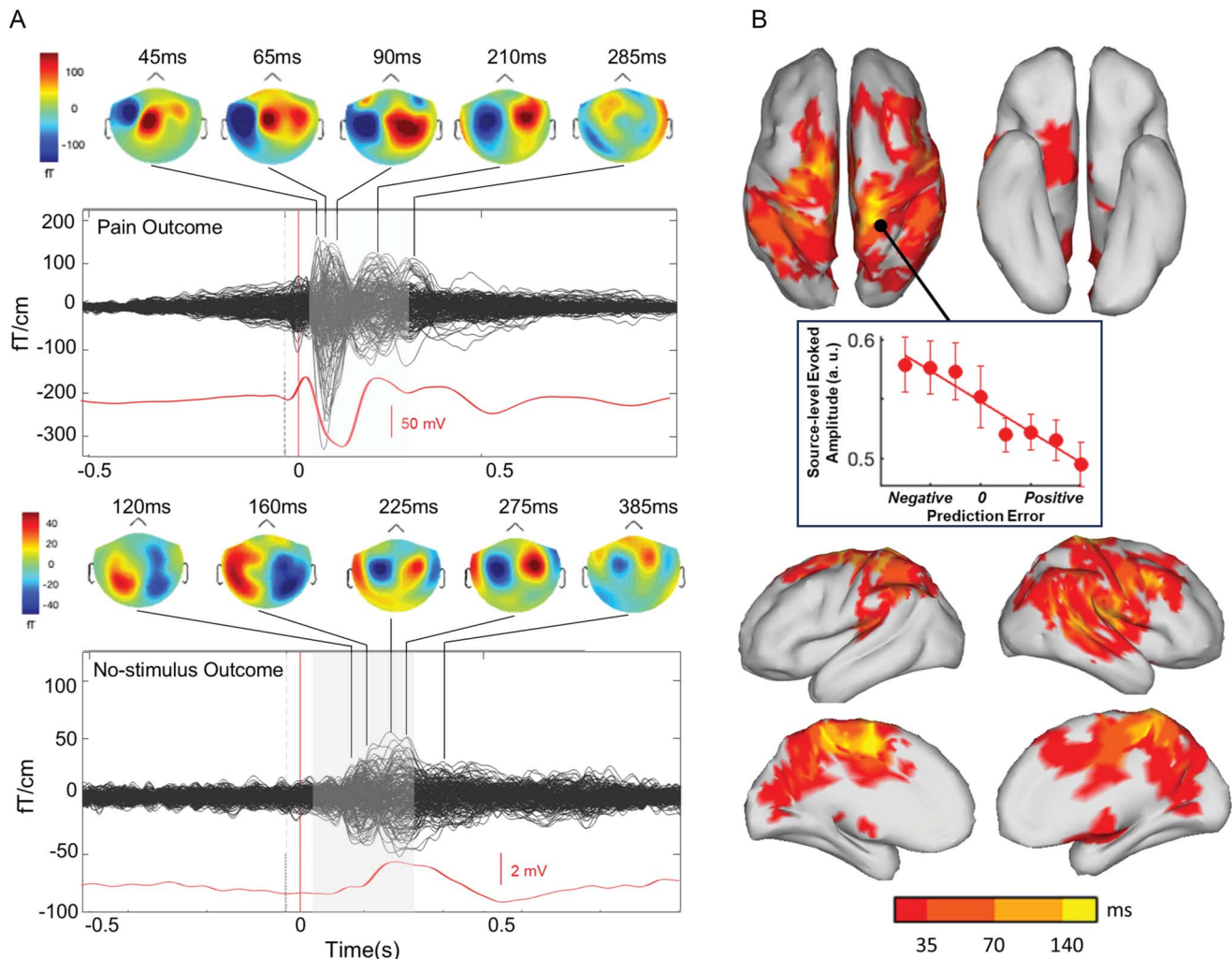


Figure 2. Model-derived aversive prediction error (PE) correlates of evoked responses. (A) Evoked responses when pain and no-stimulus outcome were delivered to right hand (RH). Top and bottom panel, respectively, show the butterfly plot of grand averaged evoked responses from MEG gradiometer channels from pain and no-stimulus delivery. The data sensor topographies derived from the peaks at specific latencies are shown above each panel. Strength of magnetic fields is represented in femto-Tesla (fT/cm). Black vertical line denotes the outcome onset cue that is, asterisk presentation on the screen. Red vertical line denotes the onset of pain outcome. Red horizontal line plot in each panel shows event-related potentials (microvolts) from an EEG channel (Cz – Avg for pain outcome and Pz – Avg for no-stimulus outcome) depicting the positive and negative deflections. It is worth noting that no-stimulus (absence of pain) also resulted in low amplitude responses. (B) Cortical correlates of model-derived aversive PEs. Evoked activity conjunction of LH and RH trial sets (conjunction of supplementary Figs. S3 and S4, <http://links.lww.com/PAIN/C325>) correlated with reinforcement learning model-derived PE estimates at outcome delivery cluster-thresholded ($P < 0.05$, corrected for family-wise error rate (FWER), two-tailed) with cluster-defining thresholds of $P < 0.001$. The color mapping indicates the cumulative duration during which the sources were correlated between 36 and 235 milliseconds after outcome delivery (shaded region in Fig. 2A top and bottom panel). For example, yellow regions correlated for greater duration than red regions. The inset line plot shows activity profile derived from an exemplar region indicating the negative association between evoked amplitude and PEs. For pain and no-stimulus outcomes, mean responses from evoked source time series (arbitrary units) were extracted between 36 and 235 milliseconds after outcome delivery by quartiles of model-derived PE estimates. RH and LH trial sets were further averaged. Error bars represent SEM.

RH trial sets (a logical “and” operation) to identify PE signal that were independent of laterality of pain delivery.

2.9.1. Model-derived aversive prediction errors

Trial-wise evoked and oscillatory activity in canonical frequency bands (35 ms time bins) at outcome delivery were entered into a linear regression model to identify significant source clusters that correlated with trial-wise PE estimates (regressor of interest). The resulting source maps were cluster-extent thresholded ($P < 0.05$, FDR corrected, 2-tailed) with cluster-defining thresholds of $P < 0.001$ (evoked) and $P < 0.01$ (oscillatory) on voxel-wise FWER corrected images. Because model-derived PEs are intrinsically correlated to outcomes (presence or absence of pain), the resulting effect sizes were large, necessitating voxel-

wise correction and stringent cluster-defining threshold especially for the evoked analysis.

2.9.2. Axiomatic analysis of aversive prediction errors

For this analysis, the GLM model had outcome (pain = 1, no-stimulus = -1) and model-derived expectancy estimates as regressors of interest. We deployed the following 3 contrasts: (1) pain > no-stimulus (axiom#1); (2) when pain was received, activity should be higher when pain was less expected (ie, negative correlation with expected pain; axiom#2a); and (3) when pain was avoided, activity should also be higher when pain was less expected, that is, when avoidance was more expected (axiom#2b). The resulting source maps for each of the contrasts were cluster-extent thresholded ($P < 0.05$, FDR corrected, one-

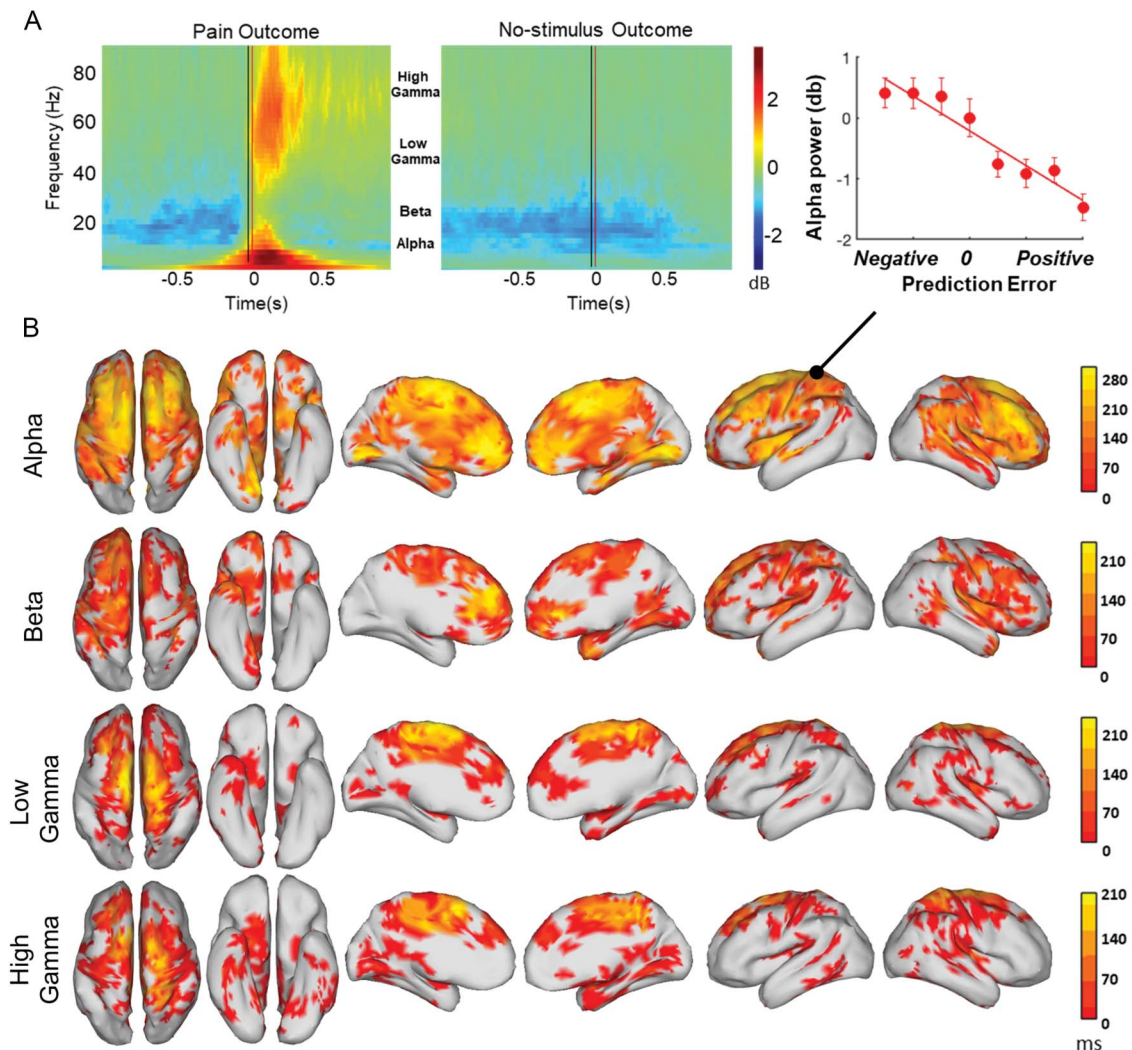


Figure 3. Model-derived aversive prediction error (PE) correlates of oscillatory responses. (A) TF responses from average source time series from the contralateral primary somatosensory cortex (S1) when pain and no-stimulus outcome were delivered to RH. Black vertical line denotes the outcome onset cue that is, asterisk presentation on the screen. Red vertical line denotes the onset of pain outcome. Frequency bands noted are alpha [8–12] Hz, beta [13–30] Hz, low gamma [30–60] Hz, and high gamma [61–90] Hz. TF maps were subjected to decibel (dB) normalization over the baseline period (1 second before the start of each trial ie, presentation of 2 visual objects in Fig. 1A). The line plot shows activity profile derived from an exemplar region indicating the negative association between alpha power and PEs. For pain and no-stimulus outcomes, mean responses from relative alpha oscillatory power (db) were extracted from 150 to 300 milliseconds after outcome delivery by quartiles of model-derived PE estimates. RH and LH trial sets were further averaged. Error bars represent SEM. (B) Cortical correlates of model-derived aversive PEs in different frequency bands. Oscillatory activity conjunction of LH and RH trial sets (conjunction of supplementary Figs. S5 and S6, <http://links.lww.com/PAIN/C325>) correlated with reinforcement learning model-derived PE estimates at outcome delivery cluster-thresholded ($P < 0.05$, corrected for family-wise error rate (FWER), two-tailed) with cluster-defining thresholds of $P < 0.01$. Sources are color coded according to the cumulative duration they were correlated between 0 and 1 second after pain onset (eg, yellow regions correlated for greater duration than red regions).

tailed) with cluster-defining thresholds of $P < 0.01$. Significant results for effect (1) would satisfy axiom#1 and significant results for effects (2) and (3) would satisfy axiom#2. While testing these axioms, we considered both the spatial overlap of clusters and temporal and spectral overlaps. For example, spatial clusters that met the axiomatic criteria were expected to be assigned the same spectral and temporal bin. Further, a second level conjunction of LH and RH trial sets ascertained that the brain correlates of PE were independent of the laterality of outcome delivery. In addition to aversive PEs, to search for regions encoding the opposite (ie, appetitive-like) PE signals, we also tested activation that fulfilled each of the reverse axioms.

2.9.3. Expectancy effects on outcome anticipation

To further identify the brain correlates of pain expectation, that is, how anticipation of outcome contributed to the generation of

aversive PEs, we analyzed the 2 seconds expectation period preceding the delivery of outcome. Trial-wise oscillatory activity at expectation onset (display of + sign on the screen after choice decision) was entered into a linear regression with trial-wise model-derived expectancy estimates (regressor of interest). The data analysis was the same as described above, except that data trials corresponded to the onset of expectation period with a time window [−1000, 3000] milliseconds, and cortical source maps exported to SPM were time-averaged from [0, 2000] milliseconds in steps of 100 milliseconds. The resulting source maps were cluster-extent thresholded ($P < 0.05$, FDR corrected, 2-tailed) with cluster-defining thresholds of $P < 0.01$.

2.9.4. Choice difficulty

To understand the neurophysiology of how pain-related learning (aversive PEs) impacted choice decisions (ie, to stay with the

current choice or switch) in the following trial, we analyzed the decision-making period. Trial-wise oscillatory activity 500 milliseconds after the appearance of 2 visual objects on the screen (circle or square in **Fig. 1A**) was entered into a linear regression with trial-wise choice difficulty estimates derived from switch/stay probability $P_{s,t}$ (regressor of interest). Choice difficulty was estimated by subtracting 0.5 from switch/stay probability $P_{s,t}$ and taking the absolute value. A choice difficulty value of 0 means that decision to stay or switch was difficult, whereas a value of 0.5 means that the decision to stay or switch was easy. The data analysis was the same as described above, except that data trials corresponded to the display of 2 visual objects with a time window of $[-1500, 1500]$ milliseconds, and cortical source maps exported to SPM were time-averaged from $[0, 500]$ milliseconds in steps of 25 milliseconds. The resulting source maps from regression analysis were cluster-extent thresholded ($P < 0.05$, FDR corrected, two-tailed) with cluster-defining thresholds of $P < 0.01$.

2.10. Anatomical labelling

In all the above analysis, to map the identified spatial clusters with relevant anatomical regions of interest, we used a standard Desikan–Killany (DK)¹⁷ atlas with the following modifications to better reflect the terminologies used in pain literature and the anatomical regions of interest (ROI). (1) Superior frontal in the DK atlas was split into dorsomedial (dmPFC) and ventromedial prefrontal cortex (vmPFC). (2) The more rostral part of the medial orbitofrontal cortex was categorized as vmPFC. (3) Frontal polar cortex was categorized as a part of vmPFC. (4) Rostral and caudal middle frontal were combined to be identified as dorsolateral prefrontal cortex (dlPFC). (5) Pars orbitalis, pars triangularis, and pars opercularis were collectively identified as ventrolateral prefrontal cortex (vlPFC). (6) Ventral diencephalon, a region in the DK volumetric atlas, was included and renamed as midbrain/diencephalon.

3. Results

3.1. Behavior

Twenty-seven healthy human participants (19–50 years old; 13 females and 14 males with a mean age of 31 ± 10 and 30 ± 5 years, respectively) completed 2 sets of 150 trials each of an instrumental pain avoidance task during MEG data collection. The probabilistically changing cue-pain reinforcements created uncertainty about outcomes and ascertained that PEs were generated throughout the task.

As expected, participants switched choices more frequently after receiving pain than no-stimulus (LH: $23 \pm 13\%$ vs $2 \pm 2\%$ of trials, RH: $23 \pm 6\%$ vs $1 \pm 1\%$ of trials, $P < 0.0001$ for both LH and RH using paired t test). The effects of pain on choice switching also decayed over time, as evidenced by logistic regressions assessing the effects of reinforcement history (pain delivered 1 through 6 trials back) on switches (**Fig. 1D**). These behavioral metrics provide evidence that participant choice selections during the task conformed to reinforcement-based learning. Refer to Supplementary Figure S1 (<http://links.lww.com/PAIN/C325>) for a full comparison of trial sets.

The computational model parameters (median values of $[\alpha_{\text{pain}}, \alpha_{\text{nopain}}, \beta] = [0.5, 0.55, 7.87]$ for the first trial set and $[0.4, 0.72, 6.98]$ for the second trial set; see Supplementary Material and Fig. S2, <http://links.lww.com/PAIN/C325>) differed between the 2 trial sets; however, significance was observed only for α_{nopain} ($P < 0.01$, Wilcoxon rank sum test). A decrease in α_{pain} and β and an increase in α_{nopain} reflected that participants choice decisions were more efficient in the second trial set compared to first.

No significant difference was found in age or any of the task behavior parameters between sexes using independent samples t test. Refer to Supplementary Table. T1 (<http://links.lww.com/PAIN/C325>).

3.2. Aversive prediction error signals

3.2.1. Evoked responses

Pain delivery elicited rapid evoked responses (**Fig. 2A**) peaking at 45, 65, and 90 milliseconds after onset, followed by broader components from 210 to 285 milliseconds. The early components mapped over the contralateral somatosensory cortex, and the later responses covered central areas bilaterally. The corresponding EEG components (recorded simultaneously with MEG) comprised the biphasic wave patterns of the P45 and N110 event-related components corresponding to early tactile responses, and a later P200 component marking pain perception. The no-stimulus outcome elicited a series of low-magnitude evoked responses, including components with later latencies beyond 200 milliseconds. Trial-wise MEG source time-series from evoked responses were regressed on to the model-derived PE estimates. Aversive PEs (negative correlation with model-derived PEs ie, greater evoked response for negative PEs than positive PEs) manifested in bilateral parietal, occipital, prefrontal, cingulate, medial temporal, and diencephalon/midbrain regions, from 36 to 235 milliseconds after outcome delivery (**Fig. 2B**).

3.2.2. Oscillatory responses

We performed the time-frequency decomposition of trial-wise MEG source time-series at outcome delivery using complex Morlet wavelets followed by decibel (dB) standardization with respect to the baseline period (1 second before the onset of the visual objects). Pain outcome elicited 4 distinct spectral responses (**Fig. 3A**): (1) a power increase or event-related synchronization in the delta $[2, 4]$ Hz and theta $[5, 7]$ Hz bands; (2) an early synchronization followed by power decrease or event-related desynchronization in the alpha $[8, 12]$ Hz and beta $[13, 30]$ Hz bands; and (3) early synchronization in the low gamma $[30, 60]$ Hz and high gamma $[61, 90]$ Hz bands. The no-stimulus condition elicited a prolonged desynchronization lasting for up to 500 milliseconds. The delta and theta bands were excluded from PE analysis because they are associated with the transient evoked responses reported above⁴¹ and were not hypothesized to be involved in pain-related learning processes.^{34,37} Trial-wise MEG source time series from oscillatory responses were regressed on to model-derived PE estimates. We found spectral components associated with aversive PEs (negative correlation with model-derived PEs ie, greater oscillatory power for negative PEs than positive PEs) manifesting within 350 milliseconds after pain onset in the alpha and beta bands, within 35 to 350 milliseconds in the low-gamma and 70 to 315 milliseconds in the high-gamma frequency ranges. These spectral correlates of PEs mapped bilaterally over cortical regions that comprised the medial/lateral prefrontal, dorsal/ventral prefrontal, medial/lateral orbitofrontal cortex (mOFC/IOFC), cingulate, insula, central somatosensory/motor, parietal and medial temporal areas, and the diencephalon/midbrain (**Fig. 3B**).

3.3. Axiomatic analysis of aversive prediction error

Although evoked responses did not yield significant spatial clusters (conjunction of supplementary Figs. S7 and S8, <http://links.lww.com/PAIN/C325>), we found spatial clusters of alpha-band oscillatory responses that fulfilled the axioms.

Aversive PE brain correlates (**Fig. 4A and 4B**) were found in the right midbrain/diencephalon (MB/DC, 140-315 ms), right ventrolateral prefrontal cortex (vlPFC, 210-420 ms), and right IOFC (140-420 ms). These regions showed enhanced alpha-band activity for pain compared to no-stimuli outcome, and activity parametrically modulated by pain expectancy (greater desynchronization for high pain expectancy), irrespective of outcome (**Fig. 4C** top and middle panel).

Further, we analyzed the pain ratings that participants provided after pain-reinforced trials in the task as a function of model-derived expectancy estimates. We found that greater pain expectations resulted in higher pain rating (**Fig. 4C** bottom panel), although the pain intensity was kept constant throughout the task based on titrated thresholds. The assimilation of pain perception towards expectation (placebo-like effect) suggests that different systems were involved in learning and assimilation of pain perception. We found this assimilation towards expectation effect was driven by the superior parietal cortex contralateral to the extremity of pain

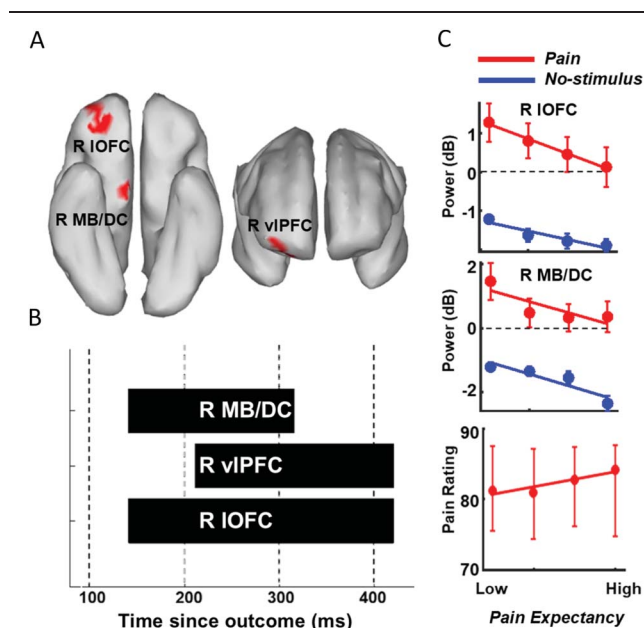


Figure 4. Aversive PE correlates satisfying axiomatic tests. (A) Spatial correlates of aversive PE: Oscillatory activity conjunction of LH and RH sets (conjunction of supplementary Figs. S9 and S10, <http://links.lww.com/PAIN/C325>) for alpha band cluster-thresholded ($P < 0.05$, FDR corrected, one-tailed) with cluster-defining thresholds of $P < 0.01$. Conjunction of LH and RH sets did not yield significant clusters in the beta or gamma bands. Three main regions are labeled. MB/DC, midbrain/diencephalon; IOFC, lateral orbito-frontal cortex; vlPFC, ventrolateral prefrontal cortex; R, right hemisphere. (B) Temporal mapping of aversive PE in each of the identified regions of interest (ROIs). Time 0 represents the onset of pain. Note: the average FWHM of the wavelet used is ~300 milliseconds in the alpha band; hence, any activity 150 milliseconds after pain onset represent true poststimulus activity arising from outcome delivery. The frequency resolution in the alpha band is ~1.5 Hz. (C) Activity profile derived from 2 select ROIs (top and middle panel) that satisfied the axioms (pain [red] > no-stimulus [blue]) and parametric variation of oscillatory activity with model-derived expectancy—on pain trials and no-stimulus trials). For pain and no-stimulus outcomes, mean responses were extracted by quartiles of model-derived pain expectancy estimates. RH and LH trial sets were further averaged. Error bars represent SEM. Effects of expectancy on pain ratings (bottom panel) from pain-reinforced trials. Average of LH and RH trial sets. Median with first and third quartiles depicted as error bars. The significance of the slope ($P = 0.0060$) was tested using a nonparametric permutation test after generating a surrogate distribution by shuffling the order of pain rating w.r.t expectancies in each participant 1000 times.

delivery (supplementary Fig. S11, <http://links.lww.com/PAIN/C325> for further details on this analysis).

3.4. Expectancy effects on outcome anticipation

We found activity correlating negatively with model-derived expectancy estimates (greater desynchronization for high pain expectancy) in several regions across the alpha to gamma frequency bands (**Fig. 5A**). In the alpha band, this activity mapped between 400 and 1900 milliseconds to the bilateral IOFC, bilateral mOFC, bilateral vlPFC, bilateral rostral ACC, bilateral fusiform, bilateral MB/DC, bilateral insula, bilateral inferior/superior/medial temporal lobe regions. In the beta band, we found correlated activity between 400 and 2000 milliseconds in bilateral parahippocampal gyrus (PH), left ventromedial prefrontal cortex (vmPFC), left IOFC, left rostral ACC, right MB/DC, left fusiform, and right lingual. In the low gamma band, this activity mapped between 1800 and 2000 milliseconds in the right MB/DC, right PH, and right entorhinal (ENT) cortex. In the high gamma band, this activity mapped between 800 and 900 milliseconds in the left MB/DC. We observed an increased correlated activity as expectations heightened towards the end of the anticipatory period (1600-2000 ms in the **Fig. 5B and 5C**). Regions encoding pain expectancy indicated enhanced activity only for low expected pain, regardless of outcome, satisfying only axiom#2 (**Fig. 5D**).

3.5. Transition from expectations to aversive prediction errors

Brain regions that satisfy the axiomatic principles of PE were expected to show a distinctive profile of activity as a function of expectancy and outcome, whereas those that track only pain expectancy were anticipated to be independent of outcome that is, violate axiom#1 (pain > no-stimuli) (**Fig. 1C**). We found that R MB/DC and R IOFC/vlPFC, which encoded aversive PEs, also encoded expectancies during the anticipatory period (**Figs. 4 and 5**). We used the fine temporal resolution of MEG signals to further track the transition of pain expectations to PEs in these regions and found that alpha band oscillations in R MB/DC and R IOFC/vlPFC gradually became more desynchronized as the outcome approached (**Fig. 6**). Moreover, we observed that the degree of alpha desynchronization was proportional to model-derived pain expectancy estimates, with greater desynchronization for higher pain expectancy. This pattern of expectancy-related desynchronization appeared to persist after the presentation of the outcome, alongside additional effects of the outcome on alpha oscillations. More specifically, delivery of the painful tactile stimulus produced increased alpha synchronization between 140 and 420 milliseconds after stimulus onset, whereas the absence of the pain stimulus had no additional effects. The observed superposition of pain-related alpha synchronization over expectancy-related alpha desynchronization in R Mb/DC and R IOFC/vlPFC aligned with our hypothesized definition of aversive PE signal (**Fig. 1C**).

3.6. Correlates of choice difficulty and decision-making

Reaction time data showed that participant made choice decisions (mouse clicks) around 600 milliseconds after the appearance of 2 visual objects (supplementary Fig. S1, <http://links.lww.com/PAIN/C325>). Hence, we included the first 500 milliseconds activity for this analysis. The reaction time also

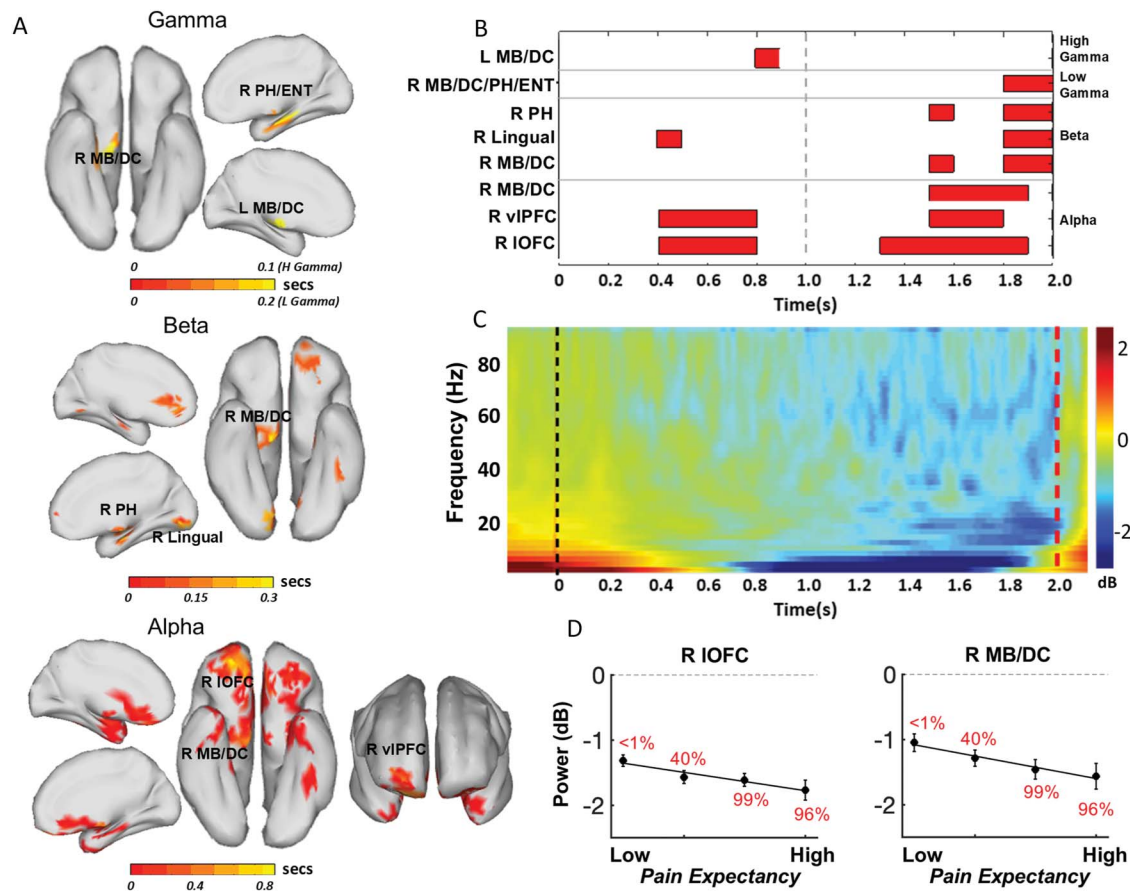


Figure 5. Expectancy effects on outcome anticipation. (A) Spatial correlates of expectancy effects during outcome anticipation. Oscillatory activity conjunction of LH and RH sets (conjunction of supplementary Figs. S12 and S13, <http://links.lww.com/PAIN/C325>) cluster-thresholded ($P < 0.05$, FDR corrected, two-tailed) with cluster-defining thresholds of $P < 0.01$. Color bar indicates the cumulative duration in secs over which the sources were correlated within the 2-second anticipatory period. Only selected regions of interest (ROIs) that exhibited prolonged correlation within each frequency band are labeled. For the gamma band, the right hemisphere correlates are from low gamma, whereas the left hemisphere correlates are from high gamma band. R indicates right and L indicates left hemispheres. Refer to text for ROI abbreviations. (B) Temporal mapping of expectancy effects in labelled ROIs in each frequency band. Time 0 represents the onset of anticipatory period that lasted for 2 seconds. The anticipatory period was terminated at outcome onset. (C) Time-frequency plot from right lateral orbitofrontal (R IOFC) during anticipatory period. Dotted vertical black line is the onset of expectation (display of + sign on the screen after choice decision), and dotted red vertical line is the termination of expectation at the onset of outcome. TF maps were subjected to decibel (dB) normalization over the baseline period (1 second before the start of each trial i.e., appearance of 2 visual objects in Fig. 1A). Note: the average FWHM of the wavelet used in the alpha, beta, low gamma, and high gamma bands were ~300 milliseconds, ~150 milliseconds, ~60 milliseconds, and <50 milliseconds, respectively. Hence, any activity before 150 milliseconds (alpha), 75 milliseconds (beta), 30 milliseconds (low gamma), and 25 milliseconds (high gamma) before outcome onset cue (2 seconds) represent true activity arising from expectations. The average frequency resolution in the alpha, beta, low gamma, and high gamma bands are ~1.5 Hz, ~3 Hz, ~6.5 Hz, and >8.5 Hz, respectively. (D) Profiles of activity derived R IOFC and right midbrain/diencephalon (R MB/DC) in the alpha band (average of LH and RH trial sets) show activity correlating negatively with model-derived expectancy estimates (greater oscillatory power for low expected pain). Mean responses were extracted by quartiles of model-derived pain expectancy estimates. Error bars represent SEM. Numbers in red indicate the percentage of trials preceded by pain outcome in the previous trial, showing low pain expectations were preceded predominantly by no-stimulus outcome, and high pain expectations were preceded predominantly by pain outcome.

correlated with choice difficulty estimates, with greater reaction times for more difficult choices (Supplementary Fig. S16, <http://links.lww.com/PAIN/C325>). We found activity correlating negatively with model-derived choice difficulty estimates (greater desynchronization for easier decisions) in the left rostral ACC and left dorsomedial prefrontal cortex (dmPFC) in the alpha band (75–125 ms), and right rostral ACC and left dmPFC in the high gamma band (225–275 ms) (Fig. 7).

There was no effect on sex on aversive PE signals, expectancy effects on outcome anticipation or choice difficulty.

4. Discussion

In the current study, we investigated when and where aversive PE signals are encoded, and how they take their roots in predictive signals preceding the outcome. We found that alpha oscillations at pain outcome fulfilled the main axiomatic requirements for an

aversive PE in MB/DC and IOFC/vIPFC. First, alpha power in these regions was increased in response to the painful tactile stimuli vs no-stimuli outcomes (axiom#1). Second, alpha power was also found to be more elevated when pain was less expected, regardless of outcome (axiom#2). Moreover, we found that these PE signals were generated by overlaying outcome-related activity (axiom#1) on top of predictive signals (axiom#2) that precede the outcome. Overall, we provide strong evidence that aversive PEs are encoded by alpha oscillations in MB/DC and IOFC/vIPFC.

4.1. Aversive prediction error signals in midbrain/diencephalon and lateral orbitofrontal cortex/ventrolateral prefrontal cortex are generated at pain onset

Observation of aversive PE signals in MB/DC and IOFC/vIPFC corresponds well with previous studies.^{24,39} Indeed, activity ascribed to the MB/DC may represent activity in midbrain areas

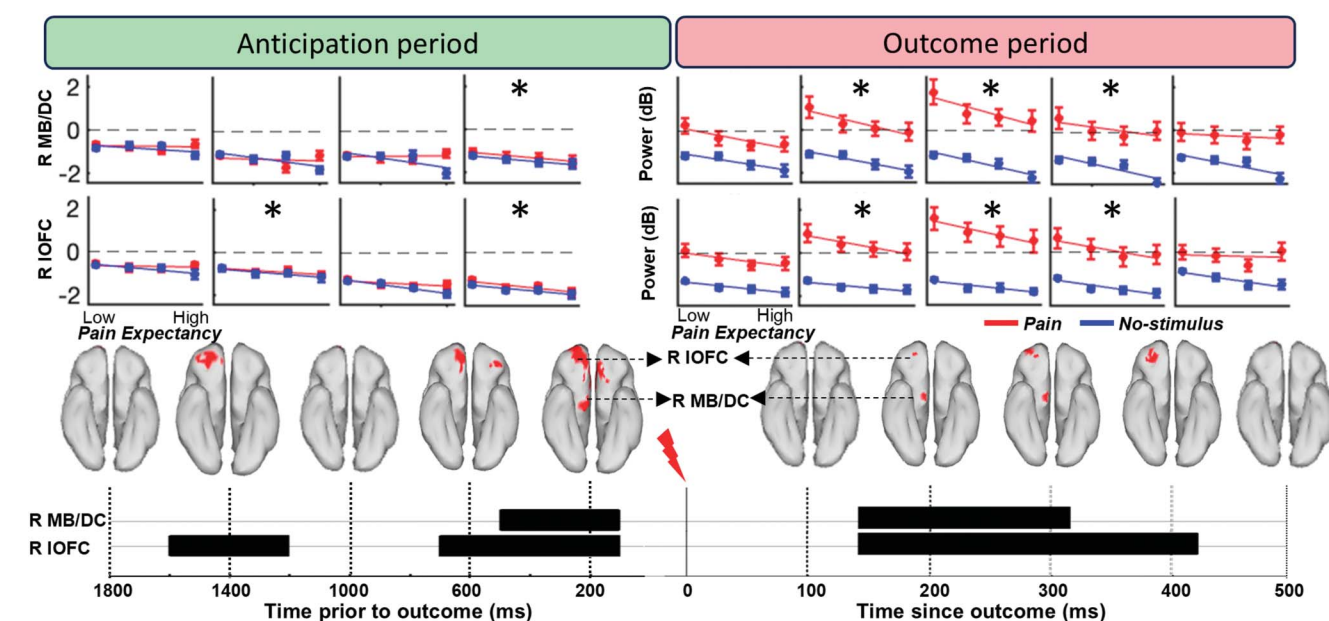


Figure 6. Transition from expectations to prediction errors (PEs). Time resolved activity in the alpha band encoding pain expectancy and aversive PE in right lateral orbitofrontal (R IOFC) and right midbrain/diencephalon (R MB/DC). These regions encoded both pain expectancy and aversive PEs in the alpha band. Horizontal bar plots (bottom row) show the time interval during which the regions conformed to the axiomatic tests indicated in Fig. 1C. Cortical maps (middle row) show the sources generating the activity at the specified time periods. Activity profiles (top row) show the alpha event-related power changes that conformed to axiomatic tests. *when axioms were satisfied with significance.

including the PAG, which has also been shown to fulfill axiomatic requirements for an aversive PE signal.^{24,39} Moreover, the MB/DC shares reciprocal connections with cortical prefrontal regions,¹³ especially the IOFC, via the superolateral medial forebrain bundle mediating reward³⁵ and aversive learning.²¹ In addition, previous fMRI studies have also observed that the IOFC/vIPFC harbors pain-specific PEs that partially fulfill axiomatic requirements.³⁹ Therefore, the current findings seem to confirm the importance of midbrain–prefrontal circuits in the generation of aversive PEs. Moreover, observation of aversive PEs in the IOFC corresponds well with a meta-analytic model of valence processing suggesting that the IOFC encodes the negative value of punishments, by comparison with the mOFC which encodes the positive value of rewards.⁹

The time course of these PE signals, which arose between 150 and 400 milliseconds after pain delivery, corresponds to the conduction velocity of nociceptive A δ (5–35 m/second) and C fibers (0.5–2 m/s). Neither the fast tactile responses (P45 in EEG) nor the slower pain-related evoked fields (P200 in EEG) fulfilled the axiomatic criteria for a PE signal. This indicates that aversive PEs are not phase-locked with pain onset, but rather depend on top-down mechanisms involved in the contextual interpretation of pain.⁴⁸ It should be noted that the increased alpha power in response to pain observed in the current study occurred at an earlier phase of stimulus processing (150–400 ms) than the pain-related desynchronization in alpha power observed in previous studies (500–1500 ms).^{22,32,33,36} This suggests that aversive PEs may occur before the pain stimulus has been fully processed.

4.2. Prediction errors originate in the predictive signals preceding pain

Prediction errors, by definition, must rely on predictive signals that precede pain onset. Remarkably, the regions found to encode PEs—MB/DC and IOFC/vIPFC—were also found to encode expectations before the delivery of outcome in a direction that

fulfilled axiom#2, that is, greater alpha desynchronization when expected pain is high,⁷ regardless of the outcome. Aversive PE signals in MB/DC and IOFC/vIPFC appear to be generated by the superimposition of outcome-related activity on top expectancy signals preceding the outcome (**Fig. 6**). We thus postulate that anticipatory alpha desynchronization in MB/DC and IOFC/vIPFC may serve to prepare the brain to quickly generate an aversive PE signal at the onset of outcome. By combining pain-related information with expectations, the MB/DC and IOFC/vIPFC clearly differentiate worse-than-expected outcomes, which are associated with increased alpha synchrony compared to baseline (top left part of the red line in **Fig. 6**), from better-than-expected outcomes, which are associated with increased alpha desynchronization compared to baseline (bottom right part of the blue line in **Fig. 6**). This PE-generating mechanism, which is initiated by the expression of predictive signals immediately before the outcome, seems to account for the quick release of signed aversive PE signals at outcome delivery.

4.3. Comparison with previous studies

Previous EEG studies on pain-related PEs have so far only examined absolute PE signals that do not discriminate between worse-than-expected and better-than-expected outcomes,⁴⁶ or did not find evidence for a signed PE signal.^{11,34} The current results cannot be compared with those previous studies because of important differences in methodology. First, most previous studies have focused on PEs related to pain intensity rather than pain occurrence.^{11,34,46} This may account for differences in the timing of responses as pain intensity-related PEs rely on processing stimulus intensity before a PE signal can be generated. Second, previous paradigms have used classical conditioning tasks with fixed reinforcement rates,^{11,14,34,46} which tend to produce little learning because of reduced associativity.²⁸ By contrast, our instrumental conditioning paradigm with variable reinforcement rate forced participants to continuously keep track

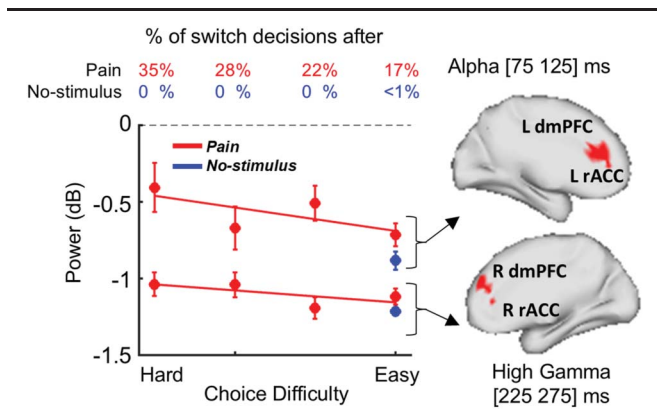


Figure 7. Correlates of choice-difficulty and decision-making. Oscillatory activity conjunction of LH and RH trial sets (conjunction of supplementary Figs. S14 and S15, <http://links.lww.com/PAIN/C325>) cluster-thresholded ($P < 0.05$, FDR corrected, two-tailed) with cluster-defining thresholds of $P < 0.01$ in the alpha and high gamma bands. Time periods next to frequency bands are relative to the appearance of 2 visual objects on the screen. The line plots show the profiles of activity derived from the labeled regions of interest (LH and RH sets averaged) modulated by choice difficulty after pain (red) and no-stimulus outcome (blue). Mean responses were extracted by quartiles of model-derived choice difficulty estimates. Error bars represent SEM. Red and blue numbers on the top, respectively, indicate the percentage of trials in which participants decided to switch choices after receiving pain and no-stimulus outcomes. As expected, choice difficulty was mostly driven by pain outcomes. No-stimulus outcomes resulted in easy choice decision to stay ($>99\%$ of trials), whereas pain outcomes resulted in difficult decision to either stay or switch. Greater choice difficulty after pain outcome led to greater proportions of switch decisions, compared to easier choice decisions.

of probabilities to choose optimally. Finally, the added spatial resolution of MEG allowed us to study signals originating from deeper structures than those reported in previous EEG studies. Altogether, our study design complemented the previous studies and allowed us to identify the sources of a signed aversive PE signal in the MB/DC and IOFC/vIPFC.

4.4. Future directions

Although, we found that increased alpha power in MB/DC and IOFC/vIPFC to fulfill the axiomatic definition of aversive PEs, it should be noted that the axioms only specify the theoretical requirements for a signal to be considered a PE. The exact neuronal processes underlying this PE-related increase in alpha power remain to be elucidated by future studies. One intriguing finding is that although *reduced* alpha power (desynchronization) is associated with *increased* pain expectation, the actual presence of pain *increases* alpha power. This pattern aligns with the PE axioms but contradicts a simpler interpretation of alpha power based solely on positive or negative valence. As mentioned earlier, anticipatory alpha desynchronization may serve to prepare the organism to process the upcoming stimulus.^{6,7,22} Increased alpha in response to pain is more difficult to interpret, but it could be related to the broadcasting of PE-related information to regions involved in decision-making to influence subsequent actions. Future studies should aim at investigating whether different neuronal subcircuits are related to the anticipatory (axiom#2) and pain-processing (axiom#1) components of the observed PE-related increases in alpha power in MB/DC and IOFC/vIPFC.

4.5. Neural oscillations of choice difficulty

Although MB/DC and IOFC/vIPFC encoded aversive PEs, we observed that bilateral dmPFC and rostral ACC encoded the

difficulty of making choices (Fig. 7). These regions are typically associated with cognitive control,^{18,49,52} possibly reflecting the increased cognitive effort needed to reach optimal decisions when faced with difficult choices between 2 options. Given that both the ACC and dmPFC receive inputs from the IOFC,³⁸ it is tempting to propose a model in which the ACC and dmPFC learn action-outcome relationships from PEs and to guide decisions.

5. Limitations

This is a whole brain association study with limited sample size and no replication. However, our sample size was based on large effect sizes produced in previous pain modulation studies with similar task design.^{4,25,39} The brain signals that correlated with our definition of PE does not imply causation, which is a general limitation inherent to all studies that infer learning-related processes from neural activity alone. Future studies that manipulate neural activity, such as through pharmacological intervention, brain stimulation, or lesion studies, are needed to prove causality. Because of the instrumental nature of our paradigm, we acknowledge the task demands were greater, which could have increased the cognitive load of the participants. In addition, because of the probabilistic nature of the task, outcome uncertainty was inherent in every trial because of which we could not test the third axiom of no surprise equivalence, an additional test that describes an aversive PE.⁴⁰ Unlike laser stimuli, the pressure pain stimuli did not specifically activate nociceptors. However, the choice of pain stimuli was deliberate because pressure pain stimuli closely resemble acute pain that a patient recovering from injury might experience because of inadvertent movement or pain resulting from touch or pressure in a patient with complex regional pain syndrome (CRPS) or fibromyalgia. Our design involved an instrumental task with self-generated expectancies, rather than cue-based expectancies that provides better experimental control. However, we believe that the effects of expectations on pain were stronger in our design likely because of greater engagement of the participants with the task.²⁷ We acknowledge that MEG has lower sensitivity and SNR to activity arising from deep structures because of the rapid decay of magnetic fields with distance. However, we accounted for this limitation by using dSPM methodology, which is a noise-normalized versions of MNE with depth-weighting to compute source signals.^{5,30}

6. Summary

Our finding provides insights into the spatial-spectral-temporal patterns of brain activity associated with pain expectancies and the generation of aversive PEs. Contrary to previous reports that predictions and PEs are encoded in different brain regions and in different frequency bands, we show that alpha oscillations encoded both pain expectancies and aversive PEs in the MB/DC and IOFC/vIPFC during very early stages of pain processing. Further, aversive PE signal in these regions were generated by simply adding outcome-related activity on top of ongoing expectancy signals. Our study advances the scientific understanding of how pain-related learning is encoded in the brain.

Conflict of interest statement

The authors have no conflict of interest to declare.

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